Solution

Since the centre is located at $(0,300)$, midway between the two foci, which are the coast guard stations, the equation is $\frac{(y-300)^2}{a^2}-\frac{(x-0)^2}{b^2}=1$. ... (1)

To determine the values of a and b, select two points known to be on the hyperbola and substitute each point in the above equation.

The point $(0,400)$ lies on the hyperbola, since it is 200 km further from Station A than from station B. $\frac{(400-300)^2}{a^2}-\frac{0}{b^2}=1$ $\frac{100^2}{a^2}=1$, $a^2=10000$. There is also a point $(x,600)$ on the hyperbola such that $600^2+x^2=(x+200)^2$.

$$\begin{aligned} 360000+x^2 &= x^2+400x+40000 \\ x &= 800 \end{aligned}$$

Substituting in (1), we have $\frac{(600-300)^2}{10000}-\frac{(800-0)^2}{b^2}=1$

$$\begin{aligned} 9-\frac{640000}{b^2} &= 1 \\ b^2 &= 80000 \end{aligned}$$

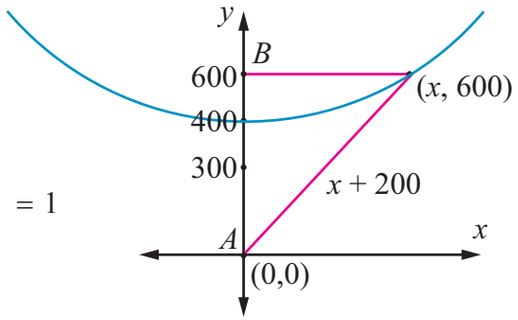


Fig. 5.67

Thus the required equation of the hyperbola is $\frac{(y-300)^2}{10000}-\frac{x^2}{80000}=1$

The ship lies somewhere on this hyperbola. The exact location can be determined using data from a third station.

Example 5.41

Certain telescopes contain both parabolic mirror and a hyperbolic mirror. In the telescope shown in figure 5.68 the parabola and hyperbola share focus F_1 which is $14m$ above the vertex of the parabola. The hyperbola's second focus F_2 is $2m$ above the parabola's vertex. The vertex of the hyperbolic mirror is $1m$ below F_1. Position a coordinate system with the origin at the centre of the hyperbola and with the foci on the y-axis. Then find the equation of the hyperbola.

Solution

Let V_1 be the vertex of the parabola and

V_2 be the vertex of the hyperbola.

$$\overline{F_1F_2} = 14-2=12m,\ 2c=12,\ c=6$$

The distance of centre to the vertex of the hyperbola is $a=6-1=5$

$$\begin{aligned} b^2 &= c^2-a^2 \\ &= 36-25=11. \end{aligned}$$

Therefore the equation of the hyperbola is $\frac{y^2}{25}-\frac{x^2}{11}=1$.

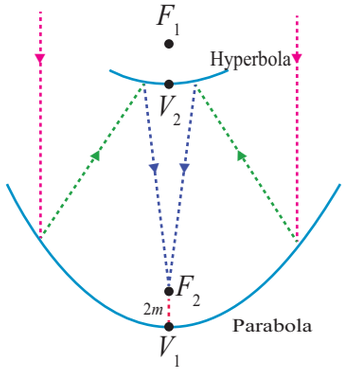


Fig. 5.68

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